1 **Design & Implementation of Human-Computer Interfaces**
NPTEL-MOOCS

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2 **L32 – Empirical Study II**

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4 **Empirical Study**

- Earlier, we learned how to evaluate usability of our designs
  - Done by experts, mostly (on prototypes)
  - In a limited scale

5 **Basic Idea**

- We also need to evaluate our system with users, to get idea on usability

6 **Basic Idea**

- When we perform a controlled experiment to collect and analyze data on user behavior, the entire process is known as *empirical research*

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Steps for Empirical Research

8 **Stages**

- Broadly, four stages
  - Identification of research question(s)
  - Determination of variables
  - Design of experiment
  - Analysis of empirical data

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Research Question

10 **Illustrative Case Study**

- Suppose we have developed a new text entry interface (a GUI to input text)

11 **Illustrative Case Study**

- We wish to collect empirical data to measure its usability
- First step – research question

12 **Research Question**

- Consider the research question

  RQ1: *Is the new technique good?*

13 **Research Question**

- We present the interface to a user and ask him/her to judge its "quality"
- We repeat the process few times (say for five users) and complete our experiment

14 **Research Question (Recorded Observation)**

15 **Research Question**

- We are dealing with observations that are difficult to interpret due to the vagueness inherent in the research question

16 **Research Question**

- Let us now "compare" our interface with other similar (text input) interfaces

17 **Research Question**

- We reformulate the research question

  RQ2: Is our text input interface better than the text input interface provided by, say, MS Word?

18 **RQ2 - Observations**

19 **Research Question**

- From these observations, we can conclude the following
  - When asked to compare, users give somewhat more concrete feedback
  - "Better" to some means more features, to others means less features
  - Some may still be confused (user #3)

20 **Research Question**

- We can do even better

21 **Research Question**

RQ3: Does the new interface let me enter text "faster" than MS Word?

22 **Task & Experiment**

- Now, we don't just ask for feedback (which may be vague and subject to interpretation)
- Now, we can give them some task (input a text) and ask for their feedback

23 **RQ3 - Observations**

24 **Research Question**

- Now easier to answer RQ3
- In fact, it's easier to analyze data avoiding subjective biases

25 **Research Question**

- Suppose we gave them the same task (same text to input)
- We recorded text entry speed (number of characters entered per minute or CPM)
- We can compare the CPM values of the five users and come to an "objective" conclusion

- Rather than relying on the users' subjective feedback on the idea of "faster"

26 **Research Question**

- So, what is there in RQ3 that makes it better than RQ1
- TWO major differences

27 **Research Question**

- Lack of ambiguity
  - We are specifying that we wish to judge quality of interface in terms of "speed of text entry"

28 **Research Question**

- Measurable quantities
  - We cannot measure interface quality – goodness is not measurable; it's always subjective
  - We have replaced unmeasurable concepts with measurable quantity – text entry speed

29 **Research Question**

- We call R3 testable and R1 as non-testable

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Testable vs non-testabale - tradeoff

31 **Tradeoff**

- Our aim - frame *testable* questions
- Problem
  - Testable questions designed to seek answer to *specific* queries
  - Such questions may lack *generalizability* to conclude about overall usability

32 **Tradeoff**

- Example – RQ3
  - Objective is to determine which interface is faster
  - Will that alone represent the interface quality?

33 **Tradeoff**

- Text entry speed is NOT the only component that determines usability of an interface

34 **Tradeoff**

- There may be other aspects of the interface that determines usability
  - Number of features supported
  - Learnability
  - Error rate
  - …

35 **Tradeoff**

- RQ3 answers only one aspect of overall usability (efficiency)

36 **Tradeoff**

- This is in contrast to RQ1, which encompasses everything (and thus untestable since we do not know what to test)

37 **Tradeoff**

- However, if we can somehow get the answer to RQ1, we are supposed to get the true conclusion

38 **Tradeoff**

- In scientific terminology, this is known as "validity" of the research question

39 **Tradeoff**

- We can draw only specific conclusions from observations made for some research questions
- Such conclusions depend on the specific test conditions and generalization is not straightforward

40 **Tradeoff**

- The *extent* to which the observations made for a research question depends on the test condition is known as the "internal validity" of the question

41 **Tradeoff**

- The *extent* to which we can generalize the conclusions drawn from the observations is called the "external validity" of the question

42 **Tradeoff**

- A trade-off
  - We cannot frame questions that are based on generalized concepts (e.g. RQ1) - those are likely to be untestable
  - If we go for more specific questions (i.e., RQ3), we might get testable questions - however, we may not get the *true* answer

43 **Tradeoff**

- We can balance the trade-off by framing *multiple* testable questions

44 **Tradeoff**

- Let us frame few more questions for our example

45 **Tradeoff**

RQ4: *Is the error rate within one hour of use less in our interface as compared to MS Word?*

RQ5: *Does the number of features supported in our interface sufficient to perform common tasks (provide feedback on a rating scale of 1 to 5 where 1 indicates "not at all" and 5 indicates "totally agree"*)?

RQ6: *Can you remember the features easily* (*on a rating scale of 1 to 5 where 1 indicates "not at all" and 5 indicates "totally agree"*)?

46 **Tradeoff**

- We captured different aspects of usability in the questions (on error rates, subjective satisfaction and memorability)
- We now perform empirical research for each separately

- Observations will lead to conclusion on overall usability - not possible with *any one* of the research questions

47 **Tradeoff**

- There is a *positive correlation* between the testable questions and the untestable question
  - We are likely to arrive at a generalized answer for an untestable question from the specific answers to multiple testable research questions

48 **Tradeoff**

- A better approach than having only untestable question and user feedback

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Note on "Hypothesis"

50 **Basic Idea**

- "Testable research questions" are more popularly known as "research hypothesis" in the domain of behavioral research

51 **Basic Idea**

- We start with two hypotheses: *null* hypothesis and *alternative* hypothesis
  - Both originate from same testable research question

52 **Example**

- We can frame two *hypotheses* from RQ3
- H_0: O*ur design is not faster than MS Word.*
- H_1: O*ur design is faster than MS Word.*

53 **Example**

- We are no longer posing any question - '?' at the end is gone
- Apart from that, there is one important difference - a single question gave rise to two hypotheses

54 **Example**

- In H_0, called the null hypothesis, we are essentially stating that the test condition is not going to affect the outcome (judgment)
  - Typically, opposite to what we set out to establish (effect of the test condition on the observations)
- H_1, called the alternative hypothesis, is just the opposite - we are stating that test condition does affect outcome

55 **Note**

- In an empirical research, we aim to find *statistical evidence* to *refute or nullify* null hypothesis and *support* alternative hypothesis

56 **Book**

- Bhattacharya, S. (2019). Human-Computer Interaction: User-Centric Computing for Design, McGraw-Hill India
  - Print Edition: ISBN-13: 978-93-5316-804-9; ISBN-10: 93-5316-804-X
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